US ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

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BRAKING, WHEELED VEHICLES

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*This TOP supersedes TOP 2-2-608, dated 20 May 2008.

1. SCOPE

This Test Operations Procedure (TOP) provides standardized tests for evaluating wheeled vehicle braking systems.

1.1 Purpose

Braking is a basic element of automotive testing, and because of its association with personnel safety, it dictates the requirement for particularly thorough testing and evaluation of wheeled vehicle braking systems to assure dependability and effectiveness under all conditions.

Major factors considered in the evaluation of wheeled vehicle braking systems are stopping and grade holding ability, vehicle stability and control when applying brakes, and individual braking system component endurance under various operational conditions.

The use of advanced technologies in the design of vehicle braking systems will dictate the test design necessary to ensure that the system is safe and effective. Some of these technologies include regenerative braking in hybrid vehicles, Antilock Brake Systems (ABS), and Electronic Stability Control (ESC) brake systems.

Brake testing involves not only tests on straight, level roads but tests on mountain highways that have long grades requiring many brake applications. Brake test evaluation also includes brake recovery capability following water immersion of the vehicle brakes in a fording basin.

Requirements and procedures for brake system testing are contained in documents such as the Federal Motor Vehicle Safety Standards (FMVSS) and the various Society of Automotive Engineers (SAE) procedures. This test procedure identifies pertinent procedures and requirements useful for testing vehicles designed for military operations.

Brake testing on surfaces affected by natural or climatic environmental conditions (mud, sand snow, etc.) that are not specifically addressed in this TOP should be considered on an individual basis per test item requirements and/or test environments and courses described in the test plan. The procedures provided within this document can be adapted to testing on lower friction surfaces to evaluate braking performance and stability. Test agency safety documents should be considered in the planning and execution process.

1.2 Limitations

This TOP describes tests to assess wheeled vehicle braking systems for compliance with applicable standards and to assess their operational capabilities. This TOP applies to wheeled vehicles designed for highway operation. These procedures may be used for both developmental and production tests. Vehicles designed specifically for off-highway operation that do not possess a maximum vehicle speed capability of at least 64 km/hr (40 mph) will be considered on an individual basis. Specific test parameters and criteria will be provided in test plans prepared for the specific vehicle under test.

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2. FACILITIES AND INSTRUMENTATION

2.1 Facilities and Test Courses

Item	Requirements
Level Paved Road	A straight, level, paved road with a lane width of not less than 3.7 m (12.0 ft), a longitudinal gradient ≤ 1%, and a side-to-side gradient ≤ 2%. Length of the roadway should be sufficient to allow the test vehicle, at its required payload condition, to accelerate to 96 km/hr (60 mph) (or maximum speed if lower than 96 km/hr) and then safely stop.
Longitudinal Grades	Longitudinal slopes ranging from 5- to 60- percent grade and of sufficient length to accommodate vehicles of various dimensions.
Fording Basin	A water basin of sufficient length and depth to completely submerge the service brake assemblies of all sizes of wheeled vehicles.
Mountain Highway Course	Reference Appendix A.
Downhill Roadway	A roadway ranging from 9- to 11-percent grade for a distance of approximately 3.2 km (2 mi) with a reasonably level surface at the bottom of the grade of sufficient length to permit a 64-km/hr (40-mph) brake stop.
Road Surfaces of Various Friction Levels	SAE J46[1**] describes the split friction surface, the changing friction surface, and the lane change test surface.
Curved Roadway	Low friction, 152- m (500-ft) radius curved roadway as described in paragraph 12 of SAE J1626[2].
Hilly Course, Paved	Paved course with grades of less than 11% that allow moderate to high road speeds, e.g., Harford Loop roadway in close proximity to Aberdeen Test Center (ATC) and the Mountain Test Course in Jennerstown, PA (reference Appendix A).
Hilly Course, Off-road	Cross-country; moderate to rough native soil and stone with grades less than 30%; e.g., ATC Churchville Test Area (CTA) course B. Cross-countrysand, gavel, and rocks with grades less than 40%; e.g., YTC Patton Hilly Trails Course.

** Superscript numbers correspond to those in Appendix D, References.

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Hilly Course, Off-road (Continued)	Secondary; improved gravel road with grades less than 10%; e.g., ATC CTA course C and Munson Test Area (MTA) gravel course. Secondary; sand gravel and rock with grades up to 30%; e.g., Yuma Test Center (YTC) Patton Hilly Gravel course.

Note: Refer to TOP 1-1-011B for details of available test courses at ATC and YTC.

2.2 Test Instrumentation

Devices for Measuring	Permissible Measurement Uncertainty (see Note 1)
Road speed	1%
Stopping distance	1%
Wheel speed	1%
Brake apply pressure	1%
Time	1%
Deceleration	2%
Pedal effort	1%
Pedal travel	2%
Brake applications count	2 counts
Force	1%
Yaw angle	2 deg
Yaw rate	1%
Brake temperature	5° C
Engine speed	1%
Voltage	1%
Current (bi-directional transducer)*	1%
Traction battery temperatures (ventilation, coolant, module(cell))	2° C
Meteorological data:	
Ambient temperature	1° C
Humidity	3%
Wind speed	5%
Wind direction	50 mrad

*Current sign conventions are as follows:

a. Current from the motor generator to the high voltage bus is positive (+).

b. Current from the traction battery to the high voltage bus is positive (+).

c. Current returned to the traction battery during a brake event is negative (-).

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d. Current used by the traction motor(s) for propulsion is negative (-).

e. Current generated by the traction motor(s) during a braking event is positive (+).

f. Current to the resistive grid (if used) is negative (-).

g. All voltage measurements are positive (+).

Note 1: The permissible measurement uncertainty is the two-standard deviation value for normally distributed instrumentation calibration data. Thus 95% of all instrumentation calibration data readings will fall within two standard deviations from the known calibration value.

2.3 Specialized Equipment/Facilities

2.3.1 Specialized equipment required for vehicle preparation, post-test inspection and test observation are as follows:

a. Micrometer calipers (inside, outside, and dial types).

b. Surface finish gauges.

c. Torque wrench.

d. Brake shoe turntable.

e. Feeler gauge stock.

f. Tire pressure gauge.

g. Video camera (to observe wheel lock and wheel lift).

h. Temperature measuring to include plug-type thermocouples, and infrared thermometer.

2.3.2 Specialized facilities and instrumentation include the following:

a. Surface Friction Measurement Device. When required, the peak friction coefficient (skid number) of the various test course surfaces will be measured in accordance with American Society for Testing and Materials (ASTM) E 1337-19[3] using an ASTM E 1136-19[4] standard reference tire.

b. Brake Force Measurement Device. A roller dynamometer used in conjunction with appropriate software (ATC currently uses the Model BM 14200 Heavy Vehicle Brake Tester***).

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c. Brake Timing Testing Device. Link Air System Expert*** is a complete package for testing application/release timing and other properties of all pneumatic brake control systems. It has an incorporated trailer test rig to properly test trailer pneumatic systems independently. The Link Air System Expert utilizes a laptop PC-based data acquisition system and includes all of the hardware and software needed to measure performance characteristics defined in FMVSS No. 121[5].

*** The use of brand names does not constitute endorsement by the Army or any other agency of the Federal Government, nor does it imply that it is best suited for its intended application.

d. Brake Application Robot. A brake application robot (AB Dynamics RBR1500*** or equivalent) may be used in conjunction with a vehicle operator to gain precise brake applications when a specific input pressure, application force/rate, or deceleration is required.

3. REQUIRED TEST CONDITIONS

3.1 Preparation for Test

a. Review all instructional material issued with the test vehicles by the manufacturer, contractor, or government, as well as reports of previous similar tests on the same types of vehicles.

b. Select the applicable test procedures to be used based on the requirements documents and purpose of the test. Obtain copies of and review the applicable FMVSS and SAE test procedures. These procedures can be listed in the detailed test plan. Note: Brake burnishing (paragraph 4.2.1) is recommended to be completed prior to all brake testing to ensure optimal performance and provide repeatable results.

c. Prepare data collection sheets to record all pre-test information, conditions of test, test results, observations, and measurements that would be valuable in analysis and assessment.

d. Ensure that all test personnel are familiar with the required technical and operational characteristics of the item and with the required test procedures.

3.2 Test Controls

a. Prior to testing ensure that:

(1) The vehicle has been prepared and equipped in accordance with standard use and/or within the specifications presented in the test plan.

(2) The vehicle is pay loaded in accordance with the test plan.

(3) The braking system components are either in good serviceable condition or new, if required.

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(4) If vehicle stability is a concern, rollover protection and/or tractor/trailer articulation restraints are considered.

(5) The peak friction coefficient (skid number) of each test track meets the requirements of the procedural documents for both dry and wet road surfaces. The skid number should be determined in accordance with ASTM E 1337-19.

(6) Automatic adjusters are activated and will remain so for the duration of the test.

(7) Brakes are adjusted, per the vehicle manufacturer's recommended procedure, at the beginning of the test. They may only be re-adjusted where specifically indicated in the test procedure.

3.3 Restrictions

Tests are not conducted at night, during inclement weather, in congested traffic, or when the road surface may introduce a hazard to the test vehicle or other traffic on the road. Dry, unobstructed surfaces are to be used unless the test plan introduces a specific requirement. Local safety and operational procedures will be carefully followed. Desirable environmental conditions for test conduct are as follows:

a. Wind speed:	≤ 3 m/s (7 mph) average value
b. Ambient temperature:	0 ≤ T ≤ 30 °C (32 ≤ T ≤ 86 °F)
c. Humidity:	≤ 95%

4. TEST PROCEDURES

4.1 Brake System Inspection/Characterization

4.1.1 Basic Measurements/Brake System Component Inspection

Specific test and vehicle identification characteristics should be recorded prior to the start of brake testing. A thorough inspection of the entire brake system should be conducted prior to the start of brake testing with details recorded. Inspections should include individual component information (type, size, reservoir capacity, etc.) and static operational tests (low air alarm activation pressure, compressor cut-in/cut-out pressure, etc.). Data sheets provided in SAE J1626 can be used as a guide. The brake system component information and static operational test results should be compared to applicable regulations provided in FMVSS No. 105[6] or 121, Federal Motor Carrier Safety Regulation (FMCSR) 393 Subpart C[7], and/or specific vehicle requirements.

Prior to the start of brake testing the condition of the brake system should be inspected to ensure compliance with FMCSR 396 Appendix A[8]. At the completion of each test/test phase the

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vehicle brake system shall meet all inspection criteria specified in FMCSR 396 Appendix A. Additionally, brake drag which causes the brake pad or lining to increase temperature while the vehicle is moving without the service brake pedal applied should not be present throughout testing. Brake drag is a term used to describe when the brake pad or lining does not properly disengage after releasing the service brake pedal.

4.1.2 Brake Pneumatic Timing (Air and Air Over Hydraulic Brake Systems Only)

Brake timing characterizes how quickly brakes can be applied and released. Testing is conducted in accordance with FMVSS No. 121. The brake actuation and release times are determined by instrumenting the brake system with pressure transducers mounted at the brake chambers and the service glad hand. The brake apply and release times are measured with the Air System Expert test apparatus, or appropriate data acquisition system. A minimum of six brake applications should be conducted during testing and the results from each application are averaged. If a specific criterion is not presented in the test plan, the requirements of FMVSS No. 121, paragraphs S5.3.3 and S5.3.4, apply.

4.1.3 Brake Balance

Determination of a vehicle's brake balance is a method of assessing the condition of its brake system prior to the initiation of brake testing. Determining the brake balance of a vehicle is accomplished by measuring the brake force distribution to individual wheels at various input pressures.

4.1.4 Brake Force Distribution and Threshold Pressures

Brake force distribution and threshold pressure characteristics are determined by means of a Heavy Vehicle Brake Tester as described in paragraph 2.3. In lieu of a brake tester, SAE J1505[9] and J1854[10] provide road test procedures and requirements for determination of a vehicle's brake force distribution and threshold pressures.

4.1.5 Pedal Effort Characteristics

The brake pedal force versus brake apply pressure characteristics are determined by instrumenting the system to measure brake pedal apply force and the resultant brake system apply pressures. Pedal effort force is measured by means of a calibrated load cell (mounted on the brake pedal) and read-out system. The driver should apply a constant brake pressure at which time the pedal effort and resultant brake system pressures are recorded. The pedal effort should be determined at various pressure increments up to the vehicle's maximum hydraulic pressure or governed air pressure.

The criterion for pedal effort is that vehicle stopping ability be in accordance with the appropriate test document (i.e., FMVSS No. 121, FMVSS No. 105, etc.) at a pedal force designated in the test plan for each specific vehicle test program. In the absence of specific criteria, a maximum pedal force of 800 N (180 lbf) will be used per MIL-STD-1472H[11].

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4.1.6 Deceleration Characteristics

The deceleration characteristics of a subject test vehicle are measured, calculated, and presented using the following methods:

a. Average deceleration of a full brake stop is defined as the speed at brake application divided by the time from brake application until the vehicle speed reaches zero. If distance measurement instrumentation is employed for the brake stop, the average deceleration can also be calculated by the square of the speed at brake application divided by twice the stopping distance. The average deceleration can only be calculated after a brake stop has occurred.

b. Mean Fully Developed Deceleration (MFDD) eliminates the initial braking phase when the brake system develops stopping force to the vehicle's wheels as well as the portion when the pedal is being released by the operator.

(1) The MFDD for a full brake stop is defined as the difference in vehicle speed at 80 percent of the vehicle speed at brake application and 10 percent of the vehicle speed at brake application divided by the time between the vehicle achieving 80 percent and 10 percent of the speed at brake application (per ISO/TR 13487:1997[12]).

(2) The MFDD for a brake snub (a brake application that is released before the vehicle reaches a completed stop) is defined as the change in velocity divided by the time between 30 and 90 percent of the total snub time (the time between brake activation and release). This measurement procedure can also be applied in assessing brake effectiveness without conducting a full stop (per ISO/TR 13487:1997).

c. Deceleration can also be measured and evaluated for the vehicle system using a GNSS-aided inertial measurement system. Experience has shown that using inertial measurements alone does not accurately compensate for vehicle pitch angle and course grade inclination. GNSS-aided inertial measurement systems have proven to be accurate for this application.

4.2 Safety/Performance Evaluation

4.2.1 Brake Burnish

The purpose of brake burnishing is to properly seat all brake components, mate the lining/pad to the brake drum/rotor to improve contact, cure the friction material, and allow the necessary transfer of friction deposits to the brake drum or rotor. Brake burnishing is recommended to be completed prior to all brake testing to ensure optimal performance and provide repeatable results.

Friction material burnishing is accomplished by utilizing the procedure described in the test plan designated requirements document, i.e.:

a. FMVSS No. 121, paragraph 6.1.8 (air brakes).

b. FMVSS No. 105, paragraphs 7.4.1 and 7.4.2 (hydraulic brakes).

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c. SAE J46/J2536[13], paragraph 7.

d. Mountain Brake burnish procedure (Appendix B).

No brake burnish criterion is presented for the FMVSS and SAE test procedures. However, the criterion for friction material burnishing for the brake burnish procedure presented in Appendix B is that not less than 80 percent of the friction material surface area be in contact with the swept area of the rotating brake member (drum or disc).

4.2.2 Grade Holding Ability

The vehicle is parked on a dry, paved, longitudinal grade in either the ascending or descending attitude. Service and parking brake systems are engaged separately to assess their individual capability to hold the vehicle stationary. Unless otherwise specified, testing should be conducted with brake temperatures at or near ambient temperature conditions. When testing a trailer or semi-trailer, a safe disconnect (mechanical, electrical, pneumatic, and hydraulic connections) from the prime mover is maintained.

Apply the service or parking brake(s) in accordance with the manufacturer's recommendation. The test sequence should, as a minimum, be as follows:

a. Position the vehicle on the designated grade.

b. Apply the service/parking brake(s).

c. Disengage the transmission.

d. If testing parking brake, with parking brake applied, release the service brake.

e. Stop the engine.

f. Wait a minimum of two minutes.

g. Observe vehicle and/or wheel movement.

h. After engine restart, descend the grade slowly while observing the ability of the service brakes to allow adequate speed control of the vehicle with the transmission disengaged.

i. Assess the ability of the service brakes to stop the vehicle on the grade after slowly descending the grade with the transmission disengaged.

The criterion for grade holding ability for both service and parking brake systems of wheeled vehicles is that each system, independent of the other, hold the vehicle stationary in both ascending and descending attitudes with the engine off and transmission disengaged on the maximum longitudinal grade over which the vehicle is required to operate. Additionally, the

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service brakes should provide adequate speed control and stopping ability when descending the grade in the forward and reverse directions with the transmission disengaged.

To determine the longitudinal grade parking brake performance of trucks, trailers, and buses intended for highway use only, the test procedures and requirements presented in SAE J360[14], J1452[15], and J293[16] can be utilized.

Note: The service brakes must overcome additional driveline torque applied to the wheels when the transmission is in gear. The driveline torque is multiplied if the vehicle platform is equipped with a two-speed transfer case and is operating in low range. Testing to determine the ability of the service brakes to stop the vehicle when descending longitudinal grades, in the forward and reverse directions, with the transmission in gear should be considered based on vehicle classification and mission profile.

4.2.3 Brake Effectiveness

Brake stopping distances are obtained from 32 and 64 km/hr (20 and 40 mph) and from additional road speeds if specifically requested. Stopping distances are measured over the input pressure range up to the point of wheel locking. A minimum of four stops, two in each direction of travel, is recommended at each application pressure and road speed. The initial brake temperature prior to each brake stop, defined as the average temperature of the service brakes on the hottest axle of the vehicle, shall be between 66 to 93°C (150 to 200°F). These tests will be conducted on a level, hard-surfaced roadway (with proper skid number) with the transmission in neutral and the vehicle at curb weight and at its rated payload condition.

In the absence of program specific criteria, the criteria for brake stopping ability are as follows:

a. Wheeled vehicles will be capable of making a straight-line full stop from a road speed of 32 km/hr (20 mph) within the distance specified in FMCSR 393.52.

b. For all other vehicle speeds tested hydraulically braked vehicles shall meet the stopping distance requirements specified in Table II of FMVSS No. 105, and air braked vehicles, including air over hydraulic, shall meet the stopping distance requirements specified in Table II of FMVSS No. 121.

Note: The stopping distance of vehicle combinations including tractor combinations will be assessed using the loaded single-unit truck requirements.

c. During all brake stops, vehicle slew shall not exceed the limits of a roadway lane width equal to 1-1/2 times the effective width of the test vehicle. The effective width is defined as the overall width of the vehicle measured at 0.5 m (19.7 in) from the ground. Wheel lock-up restrictions are defined in paragraph 6.10 of FMVSS No. 105 and paragraph 5.3.1 of FMVSS No. 121.

For vehicles equipped with an air brake system the procedures and requirements of FMVSS No. 121, paragraph 5.3.1 will apply. For vehicles equipped with a hydraulic brake system, the procedures and requirements of FMVSS No. 105, paragraph 5.1.1 will apply. Vehicles and

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vehicle combinations operating on a public highway must be capable of the brake performance described in FMCSR 393.52.

4.2.4 Brake Recovery after Water Immersion

Wheeled vehicle braking systems will be completely submerged in water for a period of 15 to 30 minutes. After immersion, recovery is determined by making brake applications from a road speed of 32 km/hr (20 mph) at a pre-selected input pressure at 1-minute intervals. Results will be compared to the dry brake performance previously established for the vehicle.

The criterion for brake recovery is that after immersion in water for a period of 15 to 30 minutes, brake stopping ability shall have achieved complete recovery after 10 brake applications over a period of 12 minutes.

4.2.5 Trailer Breakaway

This test phase will be performed in both ascending and descending attitudes on paved, longitudinal grades. The test trailer, payloaded to its gross vehicle weight rating, will be parked on the designated grade and a safe disconnect (mechanical, electrical, pneumatic, and hydraulic connections) from the prime mover will be maintained. Local safety regulations will be followed.

Due to safety concerns associated with disconnecting a trailer from its prime mover on grades of increasing percentage, the trailer's breakaway holding capability can be determined by simulating the force provided by the grade. With the trailer positioned on a level, paved surface and its breakaway feature activated, a longitudinal force, measured by an in-line load cell, will be applied to the trailer until either the wheels slide or roll. This force is then used to calculate the grade holding capability. The trailer breakaway holding capability can also be determined using a tilt table. With the trailer positioned perpendicular to the tilt axis of the table, restraining straps with enough slack to allow the trailer to slide or roll will be used to prevent the trailer from rolling off the table. Consideration should be given to the friction surface of the table prior to testing.

The criterion for trailer breakaway holding ability is that the safety brake feature be capable of holding the vehicle stationary in both ascending and descending attitudes on the maximum grade over which the vehicle is designed to operate for a period of 30 minutes. The maximum grade will be designated in the test plan.

4.2.6 Emergency Braking (Single Point Failure)

The emergency brake system of the test vehicle will be evaluated at follows:

a. For air braked systems an emergency stopping distance test will be conducted in accordance with SAE J1626, paragraph 14. The criteria for these tests are presented in Table II of FMVSS No. 121.

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b. For hydraulic braked systems an emergency stopping distance test will be conducted in accordance with SAE J1626, paragraph 15. The criteria for these tests are presented in Table II of FMVSS No. 121, and the vehicle brake performance table located in FMCSR 393.52(d).

4.2.7 Trailer Compatibility

Statically, a braking compatibility check between a truck/trailer combination can be determined using the Heavy Vehicle Brake Tester (or equivalent) to measure threshold pressures and brake force balance.

Dynamically, the test combination can perform a series of brake snubs on a level, paved test course at a low deceleration (0.9-1.5 m/s^2 (3.0-5.0 ft/s^2) at pre-selected road speeds (i.e.,80-64 km/hr (50-40 mph)) until brake temperature stabilization occurs. Repeat at a higher deceleration (1.5-2.1 m/s^2 (5.0-7.0 ft/s^2). Brake temperature balance provides a measure of compatibility.

4.2.8 Full-Effort Braking (maximum safe speed)

Full-effort brake stops will be conducted in the forward vehicle direction on a dry, level, paved surface at 16-km/hr (10 mph) road speed increments (maximum) over a speed range span of 32 km/hr (20 mph) to maximum vehicle speed (or the highest speed where safe full-effort braking can be achieved). For air and air-over-hydraulic brake systems, full-effort is defined as a brake application in which the treadle valve pressure in any of the valve's output circuit reaches 586 kilopascal (kPa) (85 pounds per square inch (psi)) within 0.2 seconds after the application is initiated. For hydraulic brake systems, full-effort means a brake application in which the force on the brake pedal reaches 622 Newtons (N) (150 pounds force (lbf)) within 0.3 seconds after the application is initiated. The initial brake temperature prior to each brake stop, defined as the average temperature of the service brakes on the hottest axle of the vehicle, shall be between 66 to 93°C (150 to 200°F). Stopping distance measurements should be normalized to the target initial vehicle speed using the procedure described in SAE J299[17]. Testing should be conducted with the vehicle at curb weight and at its rated payload condition. At each vehicle speed tested, six stops should be performed with three stops completed in each direction of travel. The number of stops at each speed may be reduced to four if each meets the stopping distance requirement with comparable results, and the field engineer observes vehicle stability and ABS (if equipped) to comply with the criteria specified in paragraph 4.2.3.

Wheel lock-up during full-effort braking can cause unfavorable handling characteristics due to a loss of directional or lateral stability that occurs if wheel lock-up is present. A loss of stability may occur on subsequent brake stops following a stop in which the vehicle remained straight and appeared stable. If wheel lock-up occurs during test runs with a vehicle equipped with ABS with an initial speed of 48 km/hr (30 mph) or greater, outside compliance with FMVSS and/or resulting in a loss of directional or lateral stability, full-effort braking should be halted, and the ABS of the vehicle inspected. If the vehicle is deemed to be performing as intended by design (e.g. ABS not included on the system), full-effort braking should be substituted with threshold braking. Threshold braking is a technique in which the vehicle operator applies the maximum amount of brake pressure to stop the vehicle in the shortest distance possible while avoiding wheel lock-up. When properly executed, this technique causes one or more tires to operate near

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the limit of traction. Because some tires may be operating well below the limit of traction due to less than optimized brake force distribution, threshold braking may produce longer braking distances than full-effort braking, and may require additional test runs for the operator to adjust in an effort to achieve optimal performance without wheel lock.

The criteria for full-effort braking are:

a. Wheeled vehicles must be capable of making full-effort brake stops in the forward direction without damage to the brake, wheel, or suspension systems at road speeds up to 96 km/hr (60 mph) (essential) or up to maximum vehicle speed (desired).

b. In the absence of program specific criteria, the requirements provided in paragraph 4.2.3 Brake Effectiveness shall be applied.

Note: Special consideration should be given to the test scope or detailed test plan of the system under test when determining the appropriate transmission setting during full-effort braking. Certain standards require that the transmission be shifted to neutral before initiating brake application. In cases where testing is to be performed with the transmission in neutral, additional full-effort brake testing with the transmission in gear (automatic transmissions) is recommended to ensure proper vehicle operation in an emergency.

4.2.9 Anti-Lock Brake System (ABS) Evaluation

The ability of the test vehicle(s) to maintain satisfactory control and provide adequate stopping capability on various road surface conditions during full-effort brake applications will be evaluated utilizing the procedures contained in SAE J46 and/or J2536. Vehicles are often equipped with multiple ABS calibrations. Each ABS calibration should be identified prior to testing and be tested appropriately based on design.

4.2.10 Regenerative Brake Performance

Hybrid electric vehicles may employ a braking system that recovers energy from the drive motors into the battery to slow the vehicle by retarding the driveline speed. This feature is referred to as regenerative braking. Depending on the vehicle design and control strategy employed, regenerative braking can range from unnoticeable to the operator to very aggressive. The ability of the vehicle to recover energy during braking events is a desirable trait and is used to help maintain the traction battery's state of charge.

Brake performance of a hybrid vehicle is evaluated using the same test procedures as a conventional drive vehicle to include brake effectiveness (paragraph 4.2.3) and full-effort braking (paragraph 4.2.8). In addition to the normal brake test parameters, the states of the traction battery, resistance grid, and motors must also be measured.

In addition, hilly test course duty cycles can be used to evaluate the capabilities of regenerative braking systems. Both paved courses (ATC's Harford Loop and the Mountain Highway Course) and off-road courses (CTA's B and C courses and the MTA gravel course) are required. The

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specific test course is traversed while maintaining a predetermined speed. The service brakes are used to control test speed. For comparative purposes, if vehicle design permits, testing is repeated with the regenerative feature turned off.

The acquired data are used to determine the power and energy developed/consumed by each major component during a single braking event or upon completion of a specific duty cycle/test course from which an energy balance can be assessed.

4.2.11 Auxiliary Braking Devices

The performance of a vehicle's auxiliary braking device (engine brake, exhaust brake, etc.) can be assessed utilizing the coast-down method described in TOP 2-2-605[18]. The vehicle is accelerated to maximum speed on a level, paved road surface before coasting to a stop. While decelerating, vehicle road speed versus time is measured and used to calculate the deceleration. This deceleration and the known vehicle mass are used to calculate the resistive force and power. Coast-down testing is conducted with the transmission in drive in system configuration both with and without the auxiliary device activated. These values, in conjunction with the engine's rated horsepower, are used to determine the percentage of rated horsepower (efficiency) absorbed by the device.

The effectiveness of an auxiliary braking device can also be evaluated during mountain brake testing (paragraph 4.3.3). The vehicle performs a minimum of one fade and one cross-country test run described for its weight class in the mountain brake test procedure presented in Appendix A, both with and without the braking device activated. The vehicle is allowed to coast down each mountain applying brake snubs when needed to maintain the prescribed speed and deceleration. Each subsequent snub can only be performed if and when the vehicle coasts to the target speed. The total number of snubs and overall brake temperature rise are recorded and used to determine the effectiveness of the braking device.

4.3 Endurance/Durability Testing

4.3.1 Structural Integrity

The test methods described in SAE J294[19] will be performed to evaluate the structural integrity of the brake system of new vehicles and existing vehicles that have undergone changes to the brake system or gross vehicle weight rating (GVWR). SAE J294 provides procedures for two tests, the Structural Endurance Test, paragraph 6.3, and the Structural Ultimate Strength Test, paragraph 6.5. The selection of the test(s) should be determined based on the modification to the vehicle/brake system, and/or maturity level of the brake system components. Service brake structural integrity criteria are presented in SAE J1404[20].

4.3.2 Low Temperature Effects

This test is conducted to ensure satisfactory operation of the moving components of the vehicle braking system under extreme cold environmental conditions. Testing is accomplished by

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actuating the braking system while the vehicle is stationary. This test is usually conducted during other cold tests (TOP 2-2-650[21] and TOP 2-2-816[22]).

The criterion for this test is that braking system components function satisfactorily at ambient air temperatures designated in the test plan for each specific vehicle without damage to seals, gaskets, or moving parts. In the absence of a specific standard, -46° C (-50°F) will be used.

4.3.3 Mountain Highway Brake System Tests

The tests outlined in Appendix A are designed to be conducted over a 40-km (25-mi) section of US Route 30 in the Jennerstown area of western Pennsylvania. The mountain brake test course is depicted in Appendix A, Figure A-1. Due to the influences associated with operating on a public roadway (e.g., traffic, traffic lights, roadside hazards), some variability in brake applications and deceleration may occur. Incidences of variability outside of parameters set forth in this standard do not invalidate full test runs. Prior to conducting Mountain Highway Brake Testing, the test vehicle must be deemed safe to operate on public roadways. At a minimum, the test vehicle must successfully complete the following tests in a closed test course environment:

a. Vehicle weight distribution and center of gravity, in accordance with TOP 2-2-801[23] and 2-2-800[24].

b. Full-effort braking, paragraph 4.2.8.

c. Structural Integrity, paragraph 4.3.1.

d. Engine/Exhaust brake performance/effectiveness, paragraph 4.2.11.

e. If vehicle dynamic stability and/or handling concerns exist, additional testing may be required at the discretion of the test engineer.

The ability of simulating the Mountain Highway Brake System Test for two and three axle single unit vehicles on the ATC Roadway Simulator is currently being studied. The test procedures, once finalized, will be included in TOP 2-2-514, Roadway Simulator[25].

4.3.3.1 Brake Fade Test.

Brake fade is identified by the inability of the vehicle to achieve average deceleration levels equivalent to those identified during effectiveness testing (cold braking). Brake fade is typically caused by an excessive buildup of heat at the friction elements of the brake system which causes the brakes to become less effective. Hydraulic brake systems can experience brake fade due to the overheating of brake fluid.

Brake fade characteristics will be determined during repeated braking operation over a downhill roadway of approximately 9- to 11-percent grade over a distance of approximately 3.2 km (2.0-mi), followed by a full stop at the bottom of the grade.

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Fade test procedures will vary for vehicles of different gross vehicle weight (GVW) classifications as shown in Appendix A, line number 13.

The criteria for brake fade are:

a. Immediately following the downgrade brake snubbing procedure, the test vehicle, traveling at the speed specified in Appendix A, must demonstrate the capability of making a full stop on level ground within the distance specified in paragraph 4.2.3.

b. Vehicle slew shall not exceed the roadway lane width limits.

As an alternative for determining the brake system fade characteristics of vehicles whose GVW is less than 4,540 kg (10,000 lb), SAE J1247[26], Simulated Mountain Brake Test Procedures, could be substituted.

4.3.3.2 High Temperature Endurance Test

A high temperature highway brake test is conducted for the purpose of evaluating the performance, fade, wear, and endurance characteristics of wheeled vehicle braking systems under conditions where elevated brake system temperatures and braking torques are a factor. The specific test procedure is outlined in Appendix A.

The criteria are:

a. After the mountain highway brake test is completed, brake component deterioration shall not have reduced vehicle stopping ability to a point below the requirements stated in paragraph 4.2.3.

b. Damage to brake, wheel, and suspension system components, such as bending, twisting, or breakage, shall not occur because of test operations.

c. At the completion of testing, the thickness of the brake linings or pads shall not be less than the applicable requirements provided in FMCSR 393.47 Paragraph d.

4.3.4 Brake Durability and Wear

The mileages accumulated during tests outlined in TOP 2-2-506A[27] will be used for brake endurance evaluation as applicable for off-highway and general operation. Various components of wheeled vehicle braking systems are subject to failure during these tests due to contamination by foreign abrasives and lubricants. Test operators will report incipient failures during the conduct of these tests for the determination of causes of specific malfunctions. During the tests all failed parts will be labeled and retained along with samples of brake fluids and contaminating elements.

Brake effectiveness tests will be performed initially, at prescribed test mileage intervals, and at the conclusion of each designated endurance phase.

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The criteria for off-highway braking system endurance are:

a. Brake components wear attributable to abrasives accumulated during normal vehicle endurance testing shall not reduce vehicle stopping ability to a point below the requirements stated in paragraph 4.2.3 over an accumulated span of 805-km (500-mi) when test course surfaces are in a wet, muddy condition. Additionally, the thickness of the brake linings or pads shall not be less than the applicable requirements provided in FMCSR 393.47 Paragraph d

b. Damage to brake, wheel, and suspension system components, such as bending, twisting, or breakage, shall not occur because of test operation

5. DATA REQUIRED

5.1 Brake System Inspection/Characterization

5.1.1 Basic Measurements/Brake System Component Inspection

a. Type of test.

b. Dates of test.

c. Vehicle identification data.

d. Vehicle weight distribution characteristics.

e. Center of Gravity (CG) location.

f. Tire description, condition, and tread depth.

g. Brake system component description (use SAE J1626, data sheet 1 as a guide).

h. Characterize features for disc brake pads, drum brake linings, and bonded lined shoes as presented in ISO International Standard, Committee Draft CD 22574[28], March 2005.

i. Brake inspection results.

j. Brake measurement results.

5.1.2 Brake Timing

a. Brake system air pressure.

b. Time required for service brake actuation and release at the service brake chambers and in a reservoir with a volume of 820 cubic cm attached to the control (service) gladhand.

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5.1.3 Brake Balance

a. Brake force / weight ratio.

b. Rolling resistance force / weight ratio.

c. Parking brake force / weight ratio.

5.1.4 Brake Force Distribution

a. Brake pedal force.

b. Primary control pressure (brake apply).

c. Brake chamber pressure.

d. Maximum brake force for each wheel.

e. Static wheel weight.

f. Dynamic wheel weight.

g. Maximum parking brake force for each wheel.

h. Threshold pressure for each wheel.

i. Crack pressure.

5.1.5 Pedal Effort Characteristics.

a. Pedal force.

b. Brake input pressure.

c. Pertinent individual system pressures, if required.

5.1.6 Deceleration Characteristics.

a. Average deceleration.

(1) Speed at brake application.

(2) Time to full stop or stopping distance.

b. MFDD for full brake stops.

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(1) Speed at brake apply

(2) Time between 80- and 10-percent of the speed at brake apply during the brake event.

c. MFDD for brake snubs.

(1) Total brake apply time (T).

(2) Speed at 0.3T and 0.9T.

5.2 Safety / Performance Evaluation

5.2.1 Brake Burnish

a. MFDD, brake application pressure, odometer reading, and brake temperatures for each of the designated stops and snubs listed in the data collection form in Appendix B, page B-2.

b. Observations of vehicle instability, brake noise, wheel locking, and brake drag events.

5.2.2 Grade Holding Ability

a. Slope grade.

b. Vehicle attitude.

c. Distance vehicle moved (note wheel roll or slide).

d. Test duration.

e. Brake control force, if required.

f. Brake temperature, if required.

g. Comments on vehicle behavior.

5.2.3 Brake Effectiveness

a. Vehicle speed.

b. Stopping distance.

c. Brake apply pressure.

d. Average deceleration.

e. Temperature of brakes.

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f. Steering correction, if required.

g. Comments on vehicle behavior or brake noise.

h. Test course skid number.

i. Brake Effectiveness Data Sheet located in Appendix C, page C-3, can be used.

5.2.4 Brake Recovery After Water Immersion

a. Initial vehicle speed.

b. Brake input pressure.

c. Number of brake application to full recovery.

d. Time of each brake application.

e. Average deceleration.

f. Distance traveled.

5.2.5 Trailer Breakaway

a. Vehicle weight.

b. Slope grade.

c. Vehicle attitude.

d. Time of test.

5.2.6 Emergency Braking (Single Point Failure)

a. Initial vehicle speed.

b. Brake input pressure.

c. Stopping distance.

d. Average deceleration.

e. Initial brake temperature.

f. Comments on vehicle behavior and wheel lock-up.

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5.2.7 Trailer Compatibility

a. Threshold pressure.

b. Brake force.

c. Vehicle speed.

d. Average deceleration.

e. Brake lining temperature.

5.2.8 Full-Effort Braking (maximum safe speed)

a. Vehicle speed and direction.

b. Stopping distance.

c. Average Deceleration.

d. Temperature of brakes.

e. Pedal force.

f. Steering correction, if required.

g. Comments on vehicle behavior and wheel lock-up.

h. Vehicle slew.

If required, data obtained from inspection of wheel, brake, and suspension system components.

5.2.9 ABS Evaluation

a. Vehicle speed.

b. Stopping distance.

c. Average deceleration.

d. Brake apply pressure.

e. Brake lining temperature.

f. Lateral deviation.

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g. Yaw rate (optional).

h. Wheel speed (optional).

i. Wheel locking location.

j. Maximum drive through speed.

k. Test course skid number.

5.2.10 Regenerative Brake Performance

a. Vehicle speed.

b. Engine speed.

c. Stopping distance.

d. Brake apply pressure.

e. Average deceleration.

f. Brake temperature.

g. Test course description (skid number, if applicable).

h. Current and voltage of specific driveline components.

i. Traction battery ventilation temperatures.

j. Traction battery coolant temperatures.

k. Traction battery module (cell) temperatures.

l. Traction battery module (cell) voltages.

m. Traction battery state of charge (SOC)

5.2.11 Auxiliary Braking Devices.

a. Vehicle weight.

b. Vehicle speed.

c. Deceleration time.

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d. Resistance-to-motion force and power.

e. Test configuration (gear range and/or auxiliary brake setting).

f. Efficiency.

g. Brake temperature rise.

h. Number of brake snubs per test run.

5.3 Endurance /Durability Testing

5.3.1 Structural Integrity

a. Vehicle speed.

b. Brake lining temperature.

c. Average deceleration.

d. Brake apply pressure.

e. Observations of any damage to brake system components.

5.3.2 Low Temperature Effects

a. Ambient temperature and humidity test conditions.

b. Observations of any damage to brake system components.

5.3.3 Mountain Highway Brake System Tests

5.3.3.1 Brake Fade Test

Data required will be in accordance with the data sheet in Appendix C, Figure C-4.

5.3.3.2 High Temperature Endurance Test

Data required will be in accordance with the data sheets in Appendix C, Figures C-1 through C-3 and C-5 through C-7.

5.3.4 Brake Durability and Wear

a. Vehicle mileage.

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b. Test course description/conditions.

c. Brake stopping distances prior to, during, and after the endurance test.

d. Extent of damage sustained by braking system components.

e. Label and retain all failed parts and samples of fluids

6. PRESENTATION OF DATA

6.1 Brake System Inspection/Characterization

a. Listing of vehicle and brake system descriptions (use SAE J1626, data sheet 1 as a guide).

b. Brake system inspection results (FMCSR 396, Appendix A)

c. Tabulation of average brake apply and release times versus axle location.

d. Tabulation of service and parking brake forces, service and parking brake force to weight ratios, and rolling resistance to weight ratio versus individual wheel location.

e. Graph depicting brake pedal effort versus input pressure.

6.2 Safety/Performance Evaluation

a. Graphs.

(1) Stopping distance and deceleration versus input pressure.

(2) Stopping distance and deceleration versus road speed.

(3) Brake recovery; stopping distance versus number of applications.

(4) Resistance-to-motion force and power versus road speed for each transmission and/or auxiliary brake condition.

b. Tabulations.

(1) Maximum grade holding ability for each brake applied and vehicle direction.

(2) Stopping distance and deceleration versus road speed.

(3) Trailer compatibility; threshold pressure and brake force for each wheel location and average maximum brake temperature for each wheel following brake snubbing procedure.

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(4) Target stopping distance, normalized stopping distance, and deceleration versus test course surface friction condition.

(5) Notation of vehicle behavior, brake noise, wheel locking, steering correction, etc.

(6) Maximum drive through speed and performance ratio achieved during low-friction surface lane change testing.

(7) Regenerative braking:

(a) Energy developed/consumed by each major component (motor generator, traction motor, traction battery, resistance grid) for each test performed.

(b) Energy recovery (ratio of the traction battery energy to the developed traction motor energy) during each braking event.

(c) Energy losses (ratio of motor generator and/or resistance grid energy change to the traction motor energy developed) during each braking event.

(d) Computed efficiencies for each vehicle configuration, speed, and/or duty cycle.

(8) Resistance-to-motion force and power versus road speed for each transmission and/or auxiliary brake setting.

(9) Power absorbed by the auxiliary brake and its percentage of available engine power (efficiency) versus road speed.

6.3 Endurance/Durability Testing

a. Graphs.

(1) Brake effectiveness from 32 and 64 km/hr; stopping distance and deceleration versus input pressure.

(2) Brake fade and endurance test temperature profile; lining temperature versus time.

b. Tabulations.

(1) Low temperature effects; listing of damage to brake system components.

(2) Brake drum/disc and lining/pad wear characteristics versus wheel location.

(3) Brake effectiveness from 32 and 64 km/hr; stopping distance and average deceleration or MFDD versus input pressure.

(4) Brake fade test; lining temperature, input pressure, vehicle speed, stopping distance, deceleration, for each fade test run.

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(5) High temperature endurance test; average and maximum lining temperatures and average input pressure for each cross-country run.

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APPENDIX A. MOUNTAIN HIGHWAY BRAKE TEST OUTLINE

1. Disassemble brake system completely.

2. Provide all new brake components.

3. Make pertinent initial brake component measurements, recording data on forms shown in Figures C-1 and C-2.

4. Install calibrated brake test instrumentation (Paragraph 2.2).

5. Reassemble brake system.

6. Adjust lining material to proper clearances according to manufacturer's specifications.

7. Bleed hydraulically actuated brake systems per manufacturer's recommended procedure.

8. Run pre-burnish effectiveness tests from 32 km/hr (20 mph).

9. Burnish brakes to achieve at least 80 percent contact between the surface areas of the lining and the drum/rotor (use procedure summarized in Appendix B).

10. Readjust brakes to recommended clearances.

11. Measure pedal force requirements over brake input pressure range.

12. Run initial effectiveness tests (reference paragraph 4.2.3), recording data on form shown as Figure C-3. Brake stop speeds will be from 32 km/hr (20 mph) and the equivalent speed of the hot stop conducted at the end of the Fade Test. Brake snubs will be conducted from 80 to 56 km/hr (50 to 35 mph) over the same range of brake input pressures as the brake stops. The first and fourth post-burnish effectiveness tests are conducted on a closed test course. The second and third post-burnish effectiveness tests require testing to be conducted in Jennerstown, PA. Effectiveness testing in Jennerstown is typically conducted on sections of U.S. route 219 where stopping a vehicle creates safety concerns. The brake snub portion of the effectiveness test provides a metric for comparison purposes during the second and third brake effectiveness test. The results from the 32 km/hr (20 mph) stops will be compared to the pre-burnish effectiveness results to determine brake performance improvements after brake burnishing and ensure no degradation.

13. Run Fade Test No.1 (east side of Laurel Mountain, Figure A-1), operating the vehicle downgrade and accelerating between brake applications as necessary to achieve required number of applications. The average speed at brake apply and release of all snubs shall be within 2 km/hr (1 mph) of the target application and release speed, respectively. The average MFDD for all snubs shall be between 2.1 and 2.7 m/s^2 (7.0 and 9.0 ft/s^2), with a target MFDD of 2.4 m/s^2 (8.0 ft/s^2).

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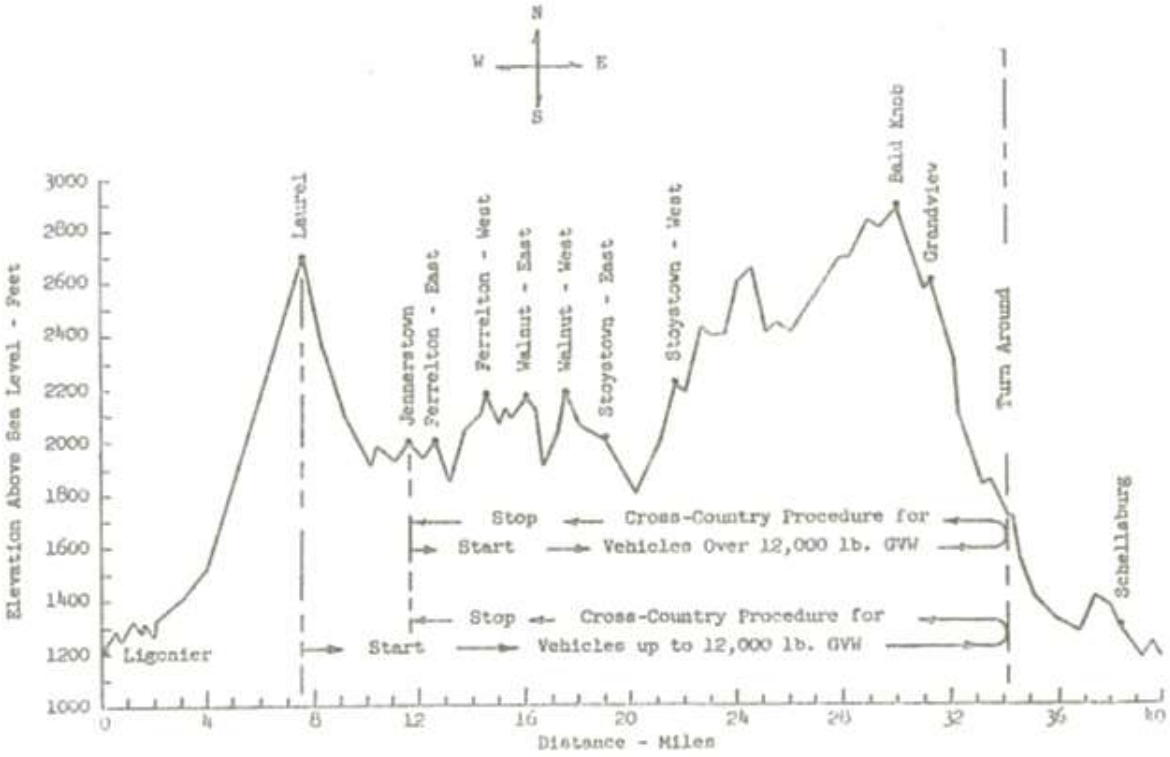


Figure A-1. Mountain Brake Test Course, Jennerstown, Pennsylvania US Route 30.

a. Snubbing and stopping rates per vehicle gross weight are as follows:

GVW up to 5,440 kg (12,000 lb):	18 applications from 64 to 32 km/hr (40 to 20 mph) at MFDD of 2.4 m/s^2 (8.0 ft/s^2) in highest gear range. One full-effort application from 64 to 0 km/hr (40 to 0 mph) on level ground at bottom of grade.
GWV 5,440 to 20,400 kg (12,000 to 45,000 lb):	30 applications from 48 to 40 km/hr (30 to 25 mph) at MFDD of 2.4 m/s^2 (8.0 ft/s^2) in highest gear range. One full-effort application from 64 to 0 km/hr (40 to 0 mph) on level ground at bottom of grade.
GVW more than 20,400 kg (45,000 lb):	25 applications from 48 to 40 km/hr (30 to 25 mph) at MFDD of 2.4 m/s^2 (8.0 ft/s^2) in lowest gear having a maximum speed of at least 56 km/hr (35 mph) at rated engine speed. One full-effort application from 48 to 0 km/hr (30 to 0 mph) on level ground at bottom of grade.

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b. Data recorded (Figure C-4) will include road speed, deceleration, input pressure, pedal travel, brake friction material temperatures, stopping distance, wheel locking, and vehicle slew.

c. Post-run data analysis will be conducted to assess average speed of brake application, speed of brake release and MFDD for all snubs. Results of the snub accuracy will be relayed to the operator for further refinements of brake application for subsequent runs.

14. Run Cross-Country Cycle No. 1:

a. Four round trips with each trip starting at 66 °C (150°F) maximum brake lining temperature.

b. All vehicles up to 5,440 kg (12,000 lb) GVW will be run in highest gear.

c. Vehicles with 5,440 kg (12,000 lb) or greater GVW will use the lowest gear when rated engine speed will provide a vehicle speed at least 8 km/hr (5 mph) more than the required snub speed. Automatic transmissions that do not allow manual selection of gears will be tested with the transmission set to 'Drive'.

d. Vehicle speed shall not exceed the posted speed limits during the cross country run.

e. GVW up to 5,440 kg (12,000 lb) - follow procedure outlined in Appendix C, Figure C-5.

f. GVW 5,440 to 20,400 kg (12,000 to 45,000 lb) - follow procedure outlined in Appendix C, Figure C-6.

g. GVW more than 20,400 kg (45,000 lb) - follow procedure outlined in Appendix C, Figure C-7.

h. Post-run data analysis will be conducted to assess average speed of brake application, speed of brake release and MFDD for all snubs. Results of the snub accuracy will be relayed to the operator for further refinements of brake applications for subsequent runs.

i. The 40-km (25-mi) section of US Route 30 can experience rapid changes in weather due to the mountainous environment. Weather conditions should be consistently monitored to avoid termination of a cross-country trip prior to completion. In the event that testing is terminated due to unfavorable weather, a cross-country trip should be deemed complete if the second hot-stop is completed prior to termination. A cross-country trip that is terminated prior to completing the second hot-stop will resume once weather conditions permit. Testing will resume at the location that testing was terminated. Due to the brake temperature profile being affected by the interruption of testing, the results from hot-stops completed after testing resumed will be used for comparison purposes at the discretion of the engineer.

15. Run Effectiveness Test No. 2 by conducting the 80 to 56 km/hr (50 to 35 mph) brake snubs over the range of brake input pressures used during Effectiveness Test No. 1.

16. Run Fade Test No. 2.

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17. Perform interim test inspection.

    a. Inspect brake system.

    b. Readjust brakes.

    c. Bleed hydraulic system.

    d. Measure lining thickness at outside of shoes/pads.

18. Run cross-country cycle No. 2 - four round trips with each trip starting at 66 ºC (150°F) brake lining temperature.

19. Run Effectiveness Test No. 3 by conducting the 80 to 56 km/hr (50 to 35 mph) brake snubs over the range of brake input pressures used during Effectiveness Test No. 1.

20. Run Fade Test No. 3.

21. Repeat paragraph 17.

22. Run Cross-Country Cycle No. 3 - four round trips with each trip starting at 66 ºC (150°F) brake lining temperature.

23. Run Fade Test No. 4.

24. Run Effectiveness Test No. 4 by conducting brake stops at 32 km/hr (20 mph) and the equivalent speed of the hot stop conducted at the end of the Fade Test over the range of brake input pressures used during Effectiveness Test No. 1. Conduct the 80 to 56 km/hr (50 to 35 mph) brake snubs over the range of brake input pressures used during Effectiveness Test No. 1.

25. Make pertinent final brake component measurements.

26. Remove instrumentation.

27. Reassemble brake system.

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APPENDIX B. MOUNTAIN BRAKE BURNISH PROCEDURE

The number of brake snubs completed at the end of each test sequence will vary depending on vehicle and brake design. The total number of brake snubs should not exceed 500 throughout the entire burnishing process. It is possible that some vehicles may not reach the target stabilized temperature specified in each test sequence. If brake temperatures stabilize below the target temperature in the test sequence, a minimum of 100 snubs should be completed at the stabilized condition, unless the result is greater than 500 total snubs. At the start of each test sequence, brake temperatures should be below 66°C (150°F) at all wheel locations. The brakes may be adjusted up to three times during the burnishing process and at the conclusion of the burnishing procedure. Brake adjustment intervals should be specified by the vehicle manufacturer prior to the start of burnishing. Brake temperatures, initial, and final input pressures should be measured throughout the entire brake burnishing procedure. If brake drag is present, as defined in paragraph 4.1.1, brake burnishing should be halted, and brake system should be repaired. If the repair process requires installation of new braking components, the new braking components should be properly burnished (as determined on a case-by-case basis).

Test Sequence 1

a. Applications to 149°C (300°F). Make brake applications per the table below at 1-mile intervals until hottest brake reaches 149°C (300°F).

Gross Vehicle Weight	MFDD		Speed at Brake Apply		Speed at Brake Release	
kg (lb)	m/s^2	ft/s^2	km/hr	mph	km/hr	mph
Up to 5,440 (12,000)	3	10	80	50	24	15
Over 5,440 (12,000)	3	10	64	40	32	20

b. Cooling Period

(1) Stop vehicle and let brakes cool to under 66°C (150°F) at all wheel locations.

(2) Record stop time.

Test Sequence 2

a. Repeat test sequence 1 step 1 until hottest brake reaches 204°C (400°F).

b. Repeat test sequence 1 step 2.

Test Sequence 3

a. Repeat test sequence 1 step 1 until hottest brake reaches 246°C (475°F).

b. Repeat test sequence 1 step 2.

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Review of Results.

a. Inspect brake shoes/pads to ensure that not less than 80 percent of the friction material surface area is in contact with the swept area of the rotating brake member (drum or disc).

b. Check for temperature balance of not more than 28°C (10°F) difference between left and right wheels on each axle. Note: Temperature balance provides indication that brakes are properly burnished. However, an underlying problem or plausible explanation can exist where brakes are properly burnished and temperature balance does not occur.

c. If the brake shoes/pads do not have 80% contact with the drum/rotor, repeat test sequence 3 until 80% contact is observed. If brake temperatures stabilize below the target temperature specified in test sequence 3, complete a minimum of 100 additional snubs at the stabilized temperature. Total snubs throughout the entire burnishing process should not exceed 500.

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AUTOMOTIVE DIRECTORATE ABERDEEN TEST CENTER, APG, MD BRAKE BURNISH DATA SHEET	Vehicle:	Date:
	USA Reg. No.:	Sheet No.:
	Brake System:	
	Vehicle Weight:	Ambient: °F
Driver: Observer:	Odometer - Start:	Stop:

Applications at _____ Intervals Stop Temperature _____ °F at Hottest Brake

Brake Apply	Time, hrs	Odom., miles	MFDD, FPS^2	Average Pressure, psi	Lining Temperature, °F												Notes
					1	2	3	4	5	6	7	8	9	10	11	12	
1																	
2																	
3																	
4																	
5																	
10																	
15																	
20																	
25																	
30																	
35																	
40																	
45																	
50																	
Hot Stop																	

Speed at Brake Apply (mph): Stopping Distance (ft):

Cooling Period: minutes, all brakes under 150°F

Notes:

Figure B-1. Brake Burnish Data Sheet.

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APPENDIX C. MOUNTAIN HIGHWAY BRAKE TEST DATA SHEETS

Automotive Directorate, Aberdeen Test Center, APG, MD
Jennerstown Brake Lining Thickness Wear Sheet

Test Vehicle:		Vehicle Weight:	lb
Brake System:	Mfgr.:		
USA Reg. No.:	Lining Material:		

Brake Location:

Location	Pre-Burnish		Post Burnish		After 1st X-Country Cycle		After 2nd X-Country Cycle		Following X-Country Completion	
	Outside	Inside	Outside	Inside	Outside	Inside	Outside	Inside	Outside	Inside
Date										
Mileage										

NOTES:

Figure C-1. Brake Lining Thickness Wear Data Sheet.

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Automotive Directorate, Aberdeen Test Center, APG, MD
Jennerstown Brake Drum Wear Sheet

Test Vehicle:				Vehicle Weight:		lb
Brake System:			Mfgr.:			
USA Reg. No.:			Drum Type:			
Drum Location:						
	Prior to Testing			After Test Completion		
Location	Opening	Center	Face	Opening	Center	Face
*						
**						
***						
Weight						
Date						
Mileage						
Drum Location:						
	Prior to Testing			After Test Completion		
Location	Opening	Center	Face	Opening	Center	Face
*						
**						
***						
Weight						
Date						
Mileage						
NOTES:						

Figure C-2. Brake Drum Wear Data Sheet.

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AUTOMOTIVE DIRECTORATE ABERDEEN TEST CENTER, APG, MD BRAKE EFFECTIVENESS DATA SHEET	Vehicle:	Date:
	Test Phase:	
	USA Reg. No.:	Sheet No.:
	Vehicle Weight:	Ambient: °F
	Odometer - Start:	Stop:
	Driver:	
	Observer:	

Target Press.		Speed at Brake Apply	Stop Distance	Apply Pressure psi		Pedal Effort	Pedal Travel	Avg. Decel	Lining Temperature degF		
psi	Seg	mph	ft	avg	max	lb	in.	ft/sec^2	min	max	Notes

AUTOMOTIVE DIRECTORATE
ABERDEEN TEST CENTER, APG, MD
LAUREL MOUNTAIN BRAKE FADE TEST, U.S. ROUTE 30, JENNERSTOWN, PA

Test Vehicle:	USA Reg. No.:	Vehicle Weight:
Brake System:		
Date:	Driver:	Observer:	Trip No.:

	Compt Disc/Run No.		Time	Odom	Counter	Ambient Temp.		Weather	Road Surface
Stop									
Start									
Total									

Vehicle Operations Requirements for Conduct of Brake Fade

__ Brake Applications from __ to __ mph at __ft/sec^2

Full Stop at bottom of the Grade from __ to 0 mph at MFDD of ____ft/sec^2

Data Required During Full Stop at the Bottom of the Grade

Road Speed: mph	Stopping Distance: ft	Avg. Deceleration: ft/sec^2
Apply Pressure:	Maximum: psi	Average: psi	Pedal Travel: in.
Comments on Vehicle Stability:		

Time, hours	Brake Apply	Pressure, psi		Pedal Travel	Notes	Time, hours	Brake Appl	Pressure, psi		Pedal Travel	Notes
		Max.	Avg					Max.	Avg		
	1						26				
	2						27				
	3						28				
	4						29				
	5						30				Medium Truck East Slope
	6						31				
	7						32				
	8						33				
	9						34				
	10						35				
	11						36				
	12						37				
	13						38				
	14						39				
	15						40				
	16						41				
	17						42				
	18				Light Truck East Slope		43				
	19						44				
	20						45				
	21						46				
	22						47				
	23						48				
	24						49				
	25				Heavy Truck East Slope		50				

Temperature, °F

	1	2	3	4	5	6	7	8	9	10	11	12
Stop												
Start												
Total												

Total Temperature Rise °F

Notes:

Figure C-4. Laurel Mountain Brake Fade Test Data Sheet.

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Automotive Directorate, Aberdeen Test Center, APG, MD
Light Truck Mountain Brake Endurance Test, U.S. Route 30, Jennerstown, PA
(for military vehicles under 12,000 lb GVW)

Test Vehicle:			USA Reg. No.:		Vehicle Weight:	lb
Brake System:						
Date:	Driver:		Observer:		Trip No.:	

Data File:	Time	Odometer	Counter	Ambient Temp	Weather	Road Surface
Stop						
Start						
Total						

LOCATION		Snub Speed, mph	No. of Snubs	Snub Counter	Clock Time	Avg. Apply Pressure	Truck Temp/ Trailer Temp (Min-max.)
East Bound on U.S. Route 30							
Laurel Mountain	TOP	40-20	12				
	BOT						
	Totals						
1st Hill East of Jennerstown		40-20	1				
Ferrelton Hill	TOP	30-20	5				
	BOT						
	Totals						
1st Hill E. Ferrelton	BOT	40-20	2				
Walnut Hill	TOP	40-20	4				
	BOT						
	Totals						
1st Hill E. Walnut	BOT	40-20	2				
Stoystown Hill	TOP	40-20	7				
	BOT						
	Totals						
Seven Mile Stretch, east bound		40-20	10				
Bald Knob	TOP	35-25	12				
	BOT						
Grandview	TOP	35-25	24				
Sugar Bear over-look							
Shot House turn							
Myers Garage							
40 mph Hot Stop:	Road Speed:	Stop Dist:		Avg Apply Press:			Deceleration:
Totals after Hot Stop							
Bald Knob to turn around	Totals						
Reverse Brake Apply @ 50% Input Pressure:							
Notes:							
West Bound on U.S. Route 30							
Seven Mile Stretch, west bound		40-20	10				
Longview Hill	TOP	40-20	3				
40 mph Hot Stop:	Road Speed:	Stop Dist:		Avg Apply Press:			Deceleration:
Totals After Hot Stop							
Stoystown Hill	TOP	40-20	10				
	BOT						
	Totals						
1st Hill W. of Stoystown		40-20	3				
Walnut Hill	TOP	40-20	6				
	BOT						
	Totals						
1st Hill W. Walnut	BOT	40-20	2				
Ferrelton Hill	TOP	40-20	4				
	BOT						
	Totals						
Ferrelton to Jennerstown		40-20	1				
Jennerstown	Totals						

Note: All snubs to be performed at a target MFDD of 8 ft/s^2. The average speed at brake apply and release of all snubs shall be within 1 mph of the target application and release speed, respectively. The average MFDD for all snubs shall be between 7 and 9 ft/s^2.

Figure C-5. Light Truck Mountain Brake Endurance Test Data Sheet.

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Automotive Directorate, Aberdeen Test Center, APG, MD
Medium Truck Mountain Brake Endurance Test, U.S. Route 30, Jennerstown, PA
(for military vehicles 12,000 to 45,000 lb GVW)

Test Vehicle:			USA Reg. No.:		Vehicle Weight:	lb
Brake System:						
Date:	Driver:		Observer:		Trip No.:	

Data File:	Time	Odometer	Counter	Ambient Temp	Weather	Road Surface
Stop						
Start						
Total						

LOCATION		Snub Speed, mph	No. of Snubs	Snub Counter	Clock Time	Avg. Apply Pressure	Truck Temp/ Trailer Temp (Min-max.)
East Bound on U.S. Route 30							
1st Hill East of Jennerstown		35-30	3				
Ferrelton Hill	TOP	30-25	10				
	BOT						
	Totals						
1st Hill E. Ferrelton	BOT	35-30	6				
Walnut Hill	TOP	30-25	12				
	BOT						
	Totals						
1st Hill E. Walnut	BOT	35-30	4				
Stoystown Hill	TOP	30-25	20				
	BOT						
	Totals						
Seven Mile Stretch, east bound		35-30	12				
Bald Knob	TOP	30-25	15				
	BOT						
Grandview	TOP	30-25	30				
Sugar Bear over-look							
Shot House turn							
Myers Garage							
40 mph Hot Stop:	Road Speed:	Stop Dist:		Avg Apply Press:		Deceleration:	
Totals after Hot Stop							
Bald Knob to turn around	Totals						
Reverse Brake Apply @ 50% Input Pressure:							
Notes:							
West Bound on U.S. Route 30							
Seven Mile Stretch, west bound		35-30	12				
Longview Hill	TOP	35-30	4				
40 mph Hot Stop:	Road Speed:	Stop Dist:		Avg Apply Press:		Deceleration:	
Totals After Hot Stop							
Stoystown Hill	TOP	30-25	20				
BOT							
	Totals						
1st Hill W. of Stoystown		35-30	6				
Walnut Hill	**TOP**	30-25	17				
	BOT						
Totals							
1st Hill W. Walnut	BOT	35-30	5				
Ferrelton Hill	TOP	30-25	10				
BOT							
Totals							
Ferrelton to Jennerstown		35-30	3				
Jennerstown	Totals						

Note: All snubs to be performed at a target MFDD of 8 ft/s^2. The average speed at brake apply and release of all snubs shall be within 1 mph of the target application and release speed, respectively. The average MFDD for all snubs shall be between 7 and 9 ft/s^2.

Figure C-6. Medium Truck Mountain Brake Endurance Test Data Sheet.

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Automotive Directorate, Aberdeen Test Center, APG, MD
Heavy Truck Mountain Brake Endurance Test, U.S. Route 30, Jennerstown, PA
(for military vehicles over 45,000 lb GVW)

Test Vehicle:			USA Reg. No.:		Vehicle Weight:	lb
Brake System:						
Date:	Driver:		Observer:		Trip No.:	

Data File:	Time	Odometer	Counter	Ambient Temp	Weather	Road Surface
Stop						
Start						
Total						

LOCATION		Snub Speed, mph	No. of Snubs	Snub Counter	Clock Time	Avg. Apply Pressure	Truck Temp/ Trailer Temp (Min-max.)
East Bound on U.S. Route 30							
1st Hill East of Jennerstown		35-30	3				
Ferrelton Hill	TOP	30-25	6				
	BOT						
	Totals						
1st Hill E. Ferrelton	BOT	35-30	4				
Walnut Hill	TOP	30-25	9				
	BOT						
	Totals						
1st Hill E. Walnut	BOT	35-30	3				
Stoystown Hill	TOP	30-25	13				
	BOT						
	Totals						
Seven Mile Stretch, east bound		35-30	6				
Bald Knob	TOP	30-25	10				
	BOT						
Grandview	TOP	30-25	25				
Sugar Bear over-look							
Shot House turn							
Myers Garage							
30 mph Hot Stop:	Road Speed:	Stop Dist:		Avg Apply Press:		Deceleration:	
Totals after Hot Stop							
Bald Knob to turn around	Totals						
Reverse Brake Apply @ 50% Input Pressure:							
Notes:							
West Bound on U.S. Route 30							
Seven Mile Stretch, west bound		35-30	6				
Longview Hill	TOP	35-30	4				
30 mph Hot Stop:	Road Speed:	Stop Dist:		Avg Apply Press:		Deceleration:	
Totals After Hot Stop							
Stoystown Hill	TOP	30-25	13				
BOT							
	Totals						
1st Hill W. of Stoystown		35-30	4				
Walnut Hill	**TOP**	30-25	11				
	BOT						
Totals							
1st Hill W. Walnut	BOT	35-30	2				
Ferrelton Hill	TOP	30-25	7				
BOT							
Totals							
Ferrelton to Jennerstown		35-30	2				
Jennerstown	Totals						

Note: All snubs to be performed at a target MFDD of 8 ft/s^2. The average speed at brake apply and release of all snubs shall be within 1 mph of the target application and release speed, respectively. The average MFDD for all snubs shall be between 7 and 9 ft/s^2.

Figure C-7. Heavy Truck Mountain Brake Test Data Sheet

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APPENDIX D. REFERENCES

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APPENDIX E. APPROVAL AUTHORITY.

DEPARTMENT OF THE ARMY
U.S. ARMY WHITE SANDS MISSILE RANGE
100 HEADQUARTERS AVENUE
WHITE SANDS MISSILE RANGE, NEW MEXICO 88002-5000

TEWS-ED

MEMORANDUM FOR SEE DISTRIBUTION

SUBJECT: Test Operations Procedure 02-2-608A Braking, Wheeled Vehicles

1. Test Operations Procedure (TOP) 02-2-608A Braking, Wheeled Vehicles has been reviewed by the U.S. Army Test and Evaluation Command (ATEC) Test Centers, the U.S. Army Operational Test Command, and the U.S. Army Evaluation Center. All comments received during the formal coordination period have been adjudicated by the preparing agency.

2. Scope of the document. This TOP provides standardized tests for evaluating wheeled vehicle braking systems.

3. This document is approved for publication and will be posted to the ASSIST website. The ASSIST website can be accessed at https://assist.dla.mil/.

4. Comments, suggestions, or questions on this document should be addressed to U.S. Army White Sand Missile Range (TEWS-ED), 100 Headquarters Avenue - Suite 231 White Sands Missile Range, NM 88002-5000; or e-mailed to usarmy.apg.atec.mbx.atec-standards@army.mil.

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